

Supplementary File: AI Tool Usage

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Two AI tools were used during the preparation of this manuscript: Claude Code (Anthropic), a command-line coding assistant, and Perplexity AI, a web-based search and reasoning tool. Neither tool was involved in the scientific content of the study. The research hypotheses, both pre-registration protocols (Batch 1 SHA `0d07a01`; Batch 2A SHA `a17c125`), the statistical framework, all experiment designs and their decision criteria, dataset selection, and interpretation of results were all performed by the author without AI assistance.

Claude Code assisted with programming and manuscript formatting. It helped draft and debug analysis scripts and handled LaTeX formatting tasks such as table layout and cross-reference management. Perplexity AI was used to research background literature, check claims against published work, and review a near-final draft for internal consistency. During the review it identified three errors that were subsequently corrected: an incorrect AUROC value in the abstract, an apparent contradiction between “immune by construction” and the magnitude correction step, and an incorrect verb in the companion paper citation.

All AI-generated outputs were reviewed and verified. The author takes full responsibility for the accuracy and integrity of the manuscript.